

FOUNDATIONS OF SURVEY DESIGN FOR OFFICIAL STATISTICS

Professional Training Manual

Statistical Domain | Beginner Level | Approximately 6 Hours

Audience: Government officials and public-sector professionals

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Proposed Table of Contents

- 1. Course Overview
- 2. Learning Objectives
- 3. Course Structure
- 4. Detailed Modules
- 5. Practical Exercises
- 6. Integrated Public-Sector Case Study
- 7. Assessment and Answer Key
- 8. Final Assignment
- 9. Glossary
- 10. Quick Reference
- 11. References
- 12. Content Verification Notes

Assumptions: beginner audience, six-hour delivery, and illustrative Indian public-sector examples unless specifically verified.

1. Course Overview

Course Purpose

Survey design is the planned process of turning an information need into a defensible statistical data-collection operation. It connects the decision to be informed with concepts, variables, population definitions, sampling, measurement, fieldwork, processing, estimation and reporting. A questionnaire is only one part of the design.

For official-statistics applications, the objective is not simply to collect many responses. The objective is to collect information that is fit for the intended use, with transparent definitions, controlled sources of error, appropriate methods and documented limitations.

Why This Subject Matters to Government Officials

Public-sector decisions may require information about households, individuals, enterprises, establishments, institutions, service users, behaviours, experiences or outcomes. Administrative records may answer some questions but not all. Surveys can fill information gaps when designed appropriately.

Design element	What can go wrong
Objective	The survey answers a different question from the decision-maker's need.
Population	Results describe the wrong group or unclear group.
Sampling frame	Eligible units are missed, duplicated or incorrectly included.
Sampling	Estimates are inefficient or cannot support the intended inference.
Measurement	Respondents interpret questions differently or cannot recall accurately.
Fieldwork	Interviewers apply inconsistent procedures.
Non-response	Respondents differ systematically from non-respondents.
Processing	Coding, editing or transformations introduce errors.
Reporting	Results lack denominators, uncertainty, definitions or limitations.

Audience, Prerequisites, Duration and Level

Audience: Government officials and public-sector professionals. **Prerequisites:** basic numeracy and familiarity with tables or percentages. **Duration:** approximately six hours including exercises. **Level:** Beginner.

Scope

This manual provides general educational guidance. Operational surveys must additionally comply with applicable departmental, legal, ethical, confidentiality, security, statistical and procurement requirements.

2. Learning Objectives

By the end of the course, learners should be able to:

1. Explain the role of surveys in producing statistical information.
2. Identify target populations, sampling frames, units of observation and variables.
3. Apply basic probability-sampling principles.
4. Compare simple random, systematic, stratified and cluster sampling.
5. Analyze major sources of sampling and non-sampling error.
6. Design clear, neutral and measurable questionnaire items.
7. Evaluate questions for ambiguity, bias, recall difficulty and respondent burden.
8. Design a basic end-to-end survey plan.

9. Evaluate non-response and coverage risks.
10. Apply survey-design principles to a realistic public-sector information problem.

3. Course Structure

Module	Title	Topics	Time	Expected outcome
1	Introduction to Surveys and Official Statistics	Survey, census, lifecycle, quality	45 min	Explain the survey system and major risks.
2	Objective and Population	Objectives, concepts, units, frames, variables	60 min	Produce a clear survey specification.
3	Sampling Fundamentals	Probability sampling and common designs	75 min	Select and justify a basic design.
4	Questionnaire and Measurement	Wording, scales, order, testing	75 min	Draft usable questionnaire items.
5	Fieldwork and Data Quality	Training, non-response, errors, controls	60 min	Plan field quality controls.
6	Processing, Analysis and Reporting	Coding, validation, weights, estimates	45 min	Connect design to responsible reporting.
—	Exercises and Assessment	Exercises, case study and assessment	60 min	Demonstrate integrated understanding.

4. Detailed Modules

Module 1: Introduction to Surveys and Official Statistics

Learning Objectives

- Define a survey and distinguish it from a census.
- Describe the survey lifecycle.
- Identify sampling and non-sampling error.
- Explain why design decisions affect data quality.

Key Concepts

A **survey** is a systematic process for collecting information from units in a defined population. Units may be persons, households, businesses, establishments, institutions or other entities.

A **census** attempts to enumerate all units in a defined population. A **sample survey** collects information from a subset and uses an appropriate design and estimation approach to make statements about the target population.

Official statistics are statistical information produced or published within an official statistical system. This course does not assign official status to any particular example or prescribe the responsibilities of a particular institution.

Census versus Sample Survey

Feature	Census	Sample survey
Coverage	Intended to cover all in-scope units	Selected units
Cost/time	Usually higher	Generally lower
Sampling error	Not applicable as a sampling component	Present in probability samples
Operational burden	High	Generally lower
Use	When complete enumeration is required or appropriate	When population estimates can be produced efficiently

Survey Lifecycle

1. Define information need.
2. Define concepts, indicators, target population and units.
3. Assess or construct the sampling frame.
4. Choose sampling design and plan sample size/allocation.
5. Design and test questionnaire.
6. Prepare fieldwork procedures and training.
7. Collect and monitor data.
8. Process, edit and validate.
9. Estimate, analyze and assess uncertainty.
10. Report, disseminate and evaluate.

Major Sources of Error

Sampling error results from observing a sample rather than the entire population. Non-sampling error includes coverage error, non-response, measurement error, interviewer effects, processing error and other operational failures. A large sample can reduce sampling variability while leaving serious non-sampling problems untouched.

Illustrative Public-Sector Example

A department wants to understand whether users can complete an online service application. The survey might measure completion, barriers, assistance used and satisfaction. Before questions are written, the team must decide whether the population is all eligible users, all attempts, or only submitted applications.

Common Mistakes and Good Practices

- Starting with a questionnaire before defining the decision problem.
- Assuming sample size alone guarantees quality.
- Ignoring frame coverage.
- Reporting only the response rate.
- Changing definitions during fieldwork without documentation.

Good practice: maintain a written survey specification, version-control instruments, pilot the complete process, document deviations, and separate observed findings from interpretation.

Short Case Study

A district office proposes the question, “Are you satisfied with our excellent service?” The wording is leading and presumes the service is excellent. A better approach defines satisfaction first and asks a neutral question with a documented response scale.

Module summary: Survey quality is an end-to-end property. The objective, population, sampling, measurement, fieldwork and analysis must form one coherent system.

Module 2: Defining the Survey Objective and Population

Learning Objectives

- Write measurable objectives.
- Define target populations and units.
- Distinguish a target population from a frame.
- Identify variables and indicators.

From Policy Question to Statistical Objective

“Is the programme working?” is too broad. A stronger objective specifies the outcome, population and reference period.
Illustrative objective: “Estimate the proportion of eligible users who completed the application during the reference month and identify the most frequently reported barriers.”

Concepts, Variables and Indicators

Term	Meaning	Illustrative example
Concept	Abstract idea to be measured	Accessibility
Variable	Characteristic recorded	Completion status
Indicator	Summary measure	Percentage completing
Unit	Entity observed	Eligible user
Reference period	Time interval covered	Previous calendar month

Target Population

The target population is the complete set of units about which the survey intends to make statements. Inclusion and exclusion rules should be sufficiently precise that two trained people would classify units consistently.

Sampling Frame

A sampling frame is a list, register, map or other operational representation from which the sample can be selected. It can differ from the target population. The design team should assess missing units, duplicates, outdated records and ineligible units.

Unit of Observation and Respondent

The unit of observation is the entity for which information is collected. The respondent is the person who supplies the information. A household can be the unit while one household member is the respondent.

Step-by-Step Specification

1. State the decision or information need.
2. Define the primary outcomes.
3. Define the target population.
4. Set inclusion and exclusion rules.
5. Choose unit of observation.
6. Specify reference period.
7. List required variables and indicators.
8. Identify and assess candidate frames.
9. Record assumptions and unresolved questions.

Common Mistakes

- Using “all citizens” when only a service-user population is relevant.
- Mixing reference periods.
- Confusing raw variables with indicators.
- Assuming an administrative register is complete without checking coverage.

Good Practice

Create a survey specification sheet before questionnaire drafting. If the team cannot agree on the population, outcome and unit, the project is not ready for fieldwork.

Module summary: Clear definitions prevent downstream confusion and determine what sample and questions are actually needed.

Module 3: Sampling Fundamentals

Learning Objectives

- Explain why sampling is used.
- Distinguish probability and non-probability sampling.
- Compare common probability designs.
- Explain the role of sample size and allocation.

Why Sample?

Sampling can reduce cost, time and respondent burden while providing population estimates. The validity of inference depends on the selection design, frame, response process and estimation—not merely on the number of observations.

Probability Sampling

In probability sampling, units have known or calculable selection probabilities. This creates a basis for design-based population inference and estimation of sampling uncertainty, subject to the design and data-quality conditions.

Simple Random Sampling

Every eligible unit in the frame has the same selection probability. A practical procedure is: assign unique IDs, determine the sample size, select IDs with an appropriate random mechanism, document the frame version and retain the selection record.

Systematic Sampling

Units are selected at a regular interval after a suitable start. A basic interval is $k=N/n$ when N frame units and n selections are planned, with actual implementation depending on rounding and the design. Periodicity in the frame can create problems if it aligns with the selection interval.

Stratified Sampling

The population is divided into strata and samples are selected within each stratum. Stratification can ensure representation of important groups and may improve precision when strata are internally more similar with respect to study variables.

Cluster Sampling

Clusters are groups such as geographic areas or institutions. Clusters may be sampled first and units then sampled within them. This can reduce travel and operational cost but can increase sampling variance when units within clusters are similar.

Non-Probability Sampling

Convenience, purposive, quota and voluntary-response approaches do not provide the same probability basis for population inference. They can be useful for exploratory or operational work, but claims of population representativeness require particular caution.

Design	Potential strength	Important caution
Simple random	Straightforward theory	Needs suitable frame; may be costly to operate.
Systematic	Efficient implementation	Periodic frame patterns can create problems.
Stratified	Subgroup representation and possible precision gains	Needs useful frame information.
Cluster	Can reduce field cost	May reduce statistical efficiency.
Convenience	Fast and inexpensive	Selection bias can be substantial.

Sample Size

Sample size influences precision but does not automatically remove bias. Planning should consider key estimates, desired precision, design, subgroup requirements, expected response and resources. For a simple proportion under

basic assumptions, an introductory planning expression is $n \approx z^2 p(1-p)/e^2$. More complex surveys may require finite-population corrections, design effects, non-response adjustments and design-aware variance methods.

Common Mistakes

- Choosing whoever is easiest to contact.
- Ignoring small but important subgroups.
- Applying a simple-random formula to a complex design without adjustment.
- Using a sample size formula without stating assumptions.
- Calling a convenience sample representative.

Short Case Study

A department needs estimates for several regions. A single national simple random sample may yield too few observations in smaller regions. Stratification or another design may be considered to improve regional coverage, followed by an estimation method consistent with the design.

Module summary: Sampling is a system of frame, selection, allocation, sample size, field logistics and estimation. These elements must be designed together.

Module 4: Questionnaire and Measurement Design

Learning Objectives

- Translate concepts into measurable questions.
- Write neutral and understandable questions.
- Select appropriate response categories.
- Sequence questions logically.
- Use pilot and cognitive testing.

Measurement

Measurement turns an abstract concept into observable information. A respondent must understand the question, retrieve the relevant information, judge an answer and communicate it using the supplied response format.

Good Question Characteristics

- One clear main idea.
- Neutral wording.
- Appropriate respondent and knowledge level.
- Defined reference period where needed.
- Relevant to the objective.
- Response options that are understandable and sufficiently comprehensive.

Avoid Double-Barrelled Questions

Poor: "Was the application process fast and easy?" Someone may find it fast but difficult. Better: ask about speed and ease separately if both are needed.

Avoid Leading Questions

Poor: "How satisfied were you with our excellent service?" Better: "Overall, how satisfied or dissatisfied were you with the service?"

Response Options

Response categories should be mutually understandable and aligned with the construct. Ordered scales, yes/no categories, frequency options and numeric fields are useful for different purposes. Avoid false precision.

Question Order

1. Introduction and eligibility.
2. Core factual questions.
3. Main outcomes.
4. Explanatory/context questions.
5. Sensitive or burdensome items where appropriate.
6. Classification/demographic items when relevant.

Translation and Accessibility

Multilingual surveys should preserve the intended concept, not merely translate words literally. Terms should be tested with the respondent groups who will use them. Accessibility, literacy, disability and mode of administration should be considered in design.

Pilot Testing

Pilot testing should cover the complete process: comprehension, routing, response categories, interview length, interviewer instructions, field logistics, technical workflow and data transmission where applicable.

Illustrative Example

A service survey asks, "How long did you wait?" If users had multiple visits, the question is ambiguous. A better instrument identifies the relevant visit and provides an appropriate time unit or categorical response.

Common Mistakes

- Jargon.
- Two concepts in one question.
- Unclear reference periods.
- Overlapping response categories.
- Excessive questionnaire length.
- Collecting variables that will never be used.

Module summary: Questionnaire quality is measurement quality. Every important question should have a reason for existing and a defined interpretation.

Module 5: Fieldwork, Non-response and Data Quality

Learning Objectives

- Explain standardized field procedures.
- Identify unit and item non-response.
- Distinguish major survey errors.
- Design basic quality controls.

Fieldwork Standardization

Field procedures should define eligibility, contact attempts, introductions, question delivery, neutral probing, recording, refusal handling, escalation and completion. Consistency reduces avoidable interviewer variation.

Interviewer Training

- Survey purpose and scope.
- Eligibility definitions.
- Question-by-question instructions.

- Neutral interviewing.
- Handling uncertainty and refusals.
- Use of paper or digital instruments.
- Applicable confidentiality/security rules.
- Practice interviews and feedback.

Non-response

Unit non-response occurs when a selected unit provides no usable interview. Item non-response occurs when one or more questions are unanswered. Non-response matters because respondents may differ from non-respondents on characteristics related to the estimates.

Response rate is useful for monitoring field performance, but it is not by itself a complete measure of non-response bias or overall survey quality.

Coverage Error

Coverage error occurs when the frame excludes eligible units, includes ineligible units or duplicates units. A service register may omit users who access a service through another channel.

Measurement Error

Measurement error is a difference between the recorded response and the value the survey intends to measure. It may arise from wording, recall, social desirability, interviewer effects or response categories.

Processing Error

Errors can also enter during coding, data entry, editing, transformation, linkage, weighting or analysis. Reproducible procedures and validation rules help identify them.

Quality-Control Checklist

- Track sample selection and disposition.
- Monitor contact outcomes and response patterns.
- Review missingness and unusual values.
- Use logical consistency checks.
- Document edits and corrections.
- Version-control instruments and datasets.
- Document deviations from the planned design.

Short Case Study

A survey has a good overall response rate but much lower response among younger users. The team should investigate differential coverage, contact methods, timing, mode and questionnaire burden. A single overall response-rate figure is not enough to establish that the estimates are unbiased.

Module summary: Quality depends on what happens after sample selection. Fieldwork must be controlled, monitored and documented.

Module 6: Processing, Analysis and Reporting

Learning Objectives

- Describe the processing chain.
- Recognize validation and coding needs.
- Explain the purpose of survey weights.
- Interpret estimates with appropriate caution.

Processing Chain

A typical chain is: receive data → validate structure → edit according to rules → code/classify → derive variables → apply design information and weights where required → estimate → quality-check → document.

Editing and Validation

Validation checks can identify impossible or inconsistent values. Examples include an age outside the permitted range, a completion date before an application date, or a dependent field containing a value when its screening question indicates “not applicable.”

Weights

A survey weight broadly represents how much population information a sampled or responding unit contributes to an estimate. In probability surveys, the base weight is related to the inverse of the selection probability. Further adjustments may be used for non-response or calibration depending on the methodology.

Weights are not a universal correction. They cannot automatically repair poor measurement, missing populations or severe operational failures.

Estimates and Denominators

Common outputs include totals, means, proportions and rates. Every percentage needs a defined denominator. “62% completed the application” is incomplete unless the report states who was included in the denominator and what period was measured.

Uncertainty

Probability samples permit estimation of sampling uncertainty under appropriate assumptions. Standard errors and confidence intervals may be used. Complex designs can require design-aware variance estimation.

Reporting Principles

- State the objective and population.
- Describe the sample design sufficiently for interpretation.
- Define indicators and denominators.
- Report relevant uncertainty and limitations.
- Explain important coverage or non-response issues.
- Separate findings from recommendations.
- Avoid causal claims that the design cannot support.

Module summary: The final statistic inherits the strengths and weaknesses of the design. Processing and reporting must remain consistent with the sample and documented assumptions.

5. Practical Exercises

Exercise 1: Survey Specification

Description: Develop the first page of a survey specification for an illustrative public-service user survey.

Input: Choose a service such as a certificate, application, helpline or online portal.

1. Write the decision the survey will support.
2. Define target population and inclusion rules.
3. Define unit and reference period.
4. List three primary indicators.
5. Identify a plausible sampling frame.
6. List two frame-coverage risks.

Expected output: One-page specification. **Evaluation:** objective 25%, population 25%, indicators 25%, frame/unit/reference period 25%.

Model guidance: A strong answer makes the population operational and ensures each indicator can be computed from collected variables.

Exercise 2: Sampling Choice

Description: A department needs estimates for four regions and wants to control travel costs.

Steps: Compare simple random, stratified and cluster sampling on representation, field cost, complexity and expected precision.

Expected output: One-page recommendation with assumptions.

Model guidance: Stratification may support regional representation; clustering may reduce field cost. The final choice depends on frame quality, required estimates and geography.

Exercise 3: Rewrite Questions

Description: Improve these illustrative items: “How satisfied are you with the department’s fast and excellent service?” and “Did you find the website easy and useful?”

Steps: Identify leading or double-barrelled wording; separate concepts; add a neutral response scale.

Model guidance: Ask satisfaction neutrally and separate speed from satisfaction; separate ease of use from usefulness if both are required.

6. Integrated Public-Sector Case Study

Background

Illustrative scenario: A state department operates a digital application service and wants evidence on completion and barriers. Administrative records capture submitted applications but provide limited information about abandoned attempts.

Problem

The department proposes a voluntary feedback form shown only after successful submission and intends to describe all applicants using the results.

Available Information

- Register of submitted applications.
- Operational records of initiated applications with some missing contact information.
- Application date and broad location.
- Proposed online questionnaire.
- Limited information about abandoned attempts.

Learner Tasks

- Define the target population.
- Identify the coverage problem.
- Propose a frame or combination of sources.
- Choose a sampling approach.
- Define key variables.
- Rewrite two leading questions.

- Identify non-response risks.
- Specify two quality controls.
- State what cannot safely be concluded.

Recommended Approach

First decide whether the population is all eligible application attempts, all submitted applications, or another clearly defined group. If abandonment is part of the objective, a frame containing only successful submissions cannot represent the abandoned group. The design should seek a frame covering the relevant attempts or explicitly limit the inference.

If a suitable frame exists, a probability sample may be selected. Stratification can be considered if regional or other subgroup estimates are important and frame information supports it. The questionnaire should measure completion status, barriers, assistance and relevant characteristics with neutral wording.

Worked Solution

Element	Illustrative solution
Objective	Estimate completion among eligible application attempts in a defined period and describe reported barriers.
Population	Eligible application attempts during the reference period, with explicit inclusion rules.
Frame	Application-attempt records if coverage and duplication are assessed.
Sampling	Illustrative regional stratified probability sample if regional estimates are required.
Variables	Completion, barrier category, assistance, effort/time measure, region and relevant respondent characteristics.
Risk	Abandoned attempts may be harder to contact and may differ systematically.
Controls	Disposition tracking, standardized contacts, validation, pilot testing and documentation.
Interpretation	Do not generalize to all citizens unless the design supports that population.

Lessons Learned

- The frame must match the objective.
- Successful-user feedback is not automatically representative of unsuccessful users.
- Questionnaire quality cannot repair a fundamentally unsuitable frame.
- Sample size alone does not establish representativeness.
- Limitations should be reported.

7. Assessment

Multiple-Choice Questions

1. What should normally be defined first?

Answer: B. Survey objective

Explanation: The objective determines what information is required.

2. A sampling frame is:

Answer: B. A list or representation from which a sample is selected

Explanation: It is the operational basis for selection.

3. Which is probability sampling?

Answer: C. Simple random sampling

Explanation: Units have defined selection probabilities.

4. A leading question is one that:

Answer: B. Nudges respondents toward an answer

Explanation: Wording can influence measurement.

5. Non-response bias is concerning when:

Answer: B. Non-respondents differ systematically from respondents

Explanation: Such differences can distort estimates.

6. What does a larger sample not automatically guarantee?

Answer: C. Absence of bias

Explanation: Coverage and measurement problems can remain.

7. Why use stratification?

Answer: B. To support subgroup representation or improve precision

Explanation: Strata can make the design more useful or efficient.

8. A double-barrelled question:

Answer: B. Contains two concepts that may receive different answers

Explanation: One response cannot reliably represent two constructs.

9. A survey weight broadly represents:

Answer: B. A unit's contribution to population estimation

Explanation: Weights are part of the estimation design.

10. A good report should include:

Answer: B. Objective, population, methodology and limitations

Explanation: Context is required for responsible interpretation.

True/False Questions

1. A large sample automatically makes a survey representative. False. Other sources of bias can remain.

2. A sampling frame can differ from the target population. True. Coverage must be assessed.

3. A grammatically correct question can still measure the wrong concept. True. Wording and construct definition matter.

4. Non-response can occur at both unit and item levels. True.

5. Weighting automatically fixes every survey error. False. It cannot repair all coverage or measurement problems.

Short-Answer Questions

1. Difference between target population and frame?

Answer: The target population is the population about which inference is intended; the frame is the operational representation used for selection.

2. Give two characteristics of a good question.

Answer: It should be clear and neutral, measure a defined concept, and use an appropriate reference period/response format.

3. What is non-response bias?

Answer: Potential distortion when non-respondents differ systematically from respondents in ways relevant to the estimates.

4. Why use stratified sampling?

Answer: To support representation of important groups and potentially improve precision.

5. Why document limitations?

Answer: To allow users to interpret estimates appropriately and understand assumptions.

Scenario-Based Questions

Scenario 1: A district surveys only successful online applicants and says 90% of "applicants" found the service easy. Identify the principal concern.

Answer: The covered population excludes unsuccessful or abandoned applicants. The frame and claimed target population do not match.

Scenario 2: “How satisfied were you with the department’s quick and convenient service?” What should change?

Answer: Remove evaluative wording and separate satisfaction, speed and convenience if each is required. Use neutral wording and appropriate response options.

Answer-Key Note

The explanations above are intentionally concise for training use. Trainers may ask learners to justify each answer using the principles in Modules 1–6.

8. Final Assignment

Objective

Design a basic survey plan for a government programme or public service.

Instructions

1. State the decision/information need.
2. Define population, inclusion rules and unit.
3. Define reference period.
4. List 5–8 variables and 3 indicators.
5. Identify a frame and coverage risks.
6. Recommend and justify a sampling approach.
7. Draft 8–12 questionnaire items.
8. Describe fieldwork and non-response controls.
9. Describe processing and validation checks.
10. Write a short reporting and limitations plan.

Deliverables: 4–6 page survey-design note plus draft questionnaire. **Suggested time:** 90–120 minutes.

Assessment Rubric

Criterion	Marks
Objective and population	15
Variables and indicators	15
Sampling design	20
Questionnaire	20
Fieldwork/non-response	10
Processing/quality	10
Reporting/limitations	5
Clarity/documentation	5
Total	100

9. Glossary

Bias: Systematic deviation between an estimate and the quantity it is intended to represent.

Census: Collection intended to cover all in-scope units.

Cluster: Group of units treated as a sampling unit at one or more stages.

Concept: Abstract idea that a survey seeks to measure.

Coverage error: Error caused by mismatch between target population and operational coverage.

Indicator: Summary measure used to describe a defined phenomenon.

Item non-response: Missing response to an individual question.

Measurement error: Difference between the recorded response and intended value.

Non-probability sample: Sample selected without known or calculable unit selection probabilities.

Non-response: Failure to obtain required information from a selected unit or item.

Population: Complete set of units relevant to the statistical objective.

Probability sampling: Sampling in which units have known or calculable selection probabilities.

Questionnaire: Structured instrument for collecting survey information.

Reference period: Time interval to which a question or measure refers.

Sampling error: Variation arising because a sample rather than the whole population is observed.

Sampling frame: Operational list, register, map or representation from which a sample is selected.

Stratification: Division of a population into strata before sample selection.

Target population: Population about which the survey intends to make statements.

Unit of observation: Entity for which information is collected.

Variable: Characteristic recorded for a unit.

Weight: Factor used in estimation to represent the contribution of sampled/responding units under the survey design.

10. Quick Reference

End-to-End Checklist

1. Define information need.
2. Write statistical objective.
3. Define concepts and indicators.
4. Define target population.
5. Define units and reference period.
6. Assess frame.
7. Select sampling design.
8. Plan sample size/allocation.
9. Draft questionnaire.
10. Pilot-test.
11. Prepare field protocols and training.
12. Plan non-response follow-up.
13. Collect and monitor.
14. Validate and process.
15. Apply design-aware estimation.

16. Assess uncertainty.
17. Report definitions and limitations.
18. Archive documentation and evaluate.

Decision Rules

Question	Rule
Need population estimates?	Use a design supporting the intended inference; probability sampling is generally the foundation for design-based inference.
Frame incomplete?	Assess coverage before selection; do not assume the frame equals the population.
Need subgroup estimates?	Consider stratification or allocation providing sufficient observations for those groups.
Question difficult or sensitive?	Test comprehension, wording, mode and burden before launch.
Low response in a subgroup?	Investigate differential non-response rather than relying only on overall response.
Can weighting fix it?	Only as part of a documented estimation strategy; weighting does not automatically repair all errors.

Common Warnings

- Do not call a convenience sample representative without justification.
- Do not equate response rate with overall quality.
- Do not hide frame exclusions.
- Do not change instrument versions without documentation.
- Do not report percentages without denominators and reference periods.
- Do not make causal claims from descriptive data unless the design supports them.

Important Formulae

For introductory planning of a proportion under simple assumptions: $n \approx z^2 p(1-p)/e^2$. Here n is initial sample size, z a critical value, p a planning proportion and e a desired margin of error. For a simple random sample, the approximate standard error of a sample proportion is $\sqrt{p\text{-hat}(1-p\text{-hat})/n}$. These are introductory formulas; finite-population corrections, design effects, weighting and complex variance estimation may be needed.

11. References

The following public sources were verified for inclusion. They are methodological references and do not imply endorsement or official affiliation with this course.

Principles and Recommendations for Population and Housing Censuses, Revision 4

United Nations Statistics Division, 2025

<https://unstats.un.org/unsd/demographic-social/census/>

Designing Household Survey Samples: Practical Guidelines

United Nations Statistics Division, 2008

<https://unstats.un.org/unsd/demographic-social/standards-and-methods/>

Capturing What Matters: Essential Guidelines for Designing Household Surveys

World Bank, 2021

<https://www.worldbank.org/en/programs/lsm/publication/capturing-what-matters>

12. Content Verification Notes

General Educational Guidance

The explanations of survey objectives, population definition, sampling, measurement, questionnaire design, fieldwork, non-response, processing, estimation and reporting are general educational guidance created for this manual.

Sections Requiring Subject-Matter Expert Review

- Specific sample-size calculations.
- Legal, privacy and confidentiality requirements.
- Data retention, sharing and security procedures.
- Department-specific classifications and definitions.
- Operational field protocols.
- Procurement and contracting requirements.
- Final estimation and variance methodology for complex designs.

Assumptions

- Beginner level and approximately six hours.
- No particular department, state, programme or sampling frame was supplied.
- Public-sector scenarios are illustrative, not descriptions of verified government operations.
- Terminology may differ from terminology prescribed by a specific department.

Information That May Vary

Approved methodologies, classifications, reference periods, field procedures, legal requirements, data governance rules, sampling frames and reporting practices can vary by department, state, statistical programme, law and survey purpose. They must be checked before operational deployment.

Source Verification

The United Nations Statistics Division publicly lists standards and methods and identifies guidance on household survey samples. Its current census resource lists the fourth revision of the Principles and Recommendations for Population and Housing Censuses. The World Bank's Living Standards Measurement Study publicly describes its household-survey guidance as covering survey design, sampling, questionnaires, CAPI, documentation and dissemination.